Write up to Presentation of: An improved method of extreme value analysis, Makkonen L., Tikanmäki M.

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Rare events occur almost in every field of our life - from finances to weather. A quantitative approach to these events is the extreme value analysis (EVA), which is already more than one century old. This article sums up a method - the VWLS method[1] -, which is not only better in its performance in respect to conventional methods, but also more objective.

I. HOW IT WORKS

An EVA is normally done in the following steps:

A. Choosing a cumulative distribution function (CDF), which fits your data

B. Selecting plotting position formula

C. Weighing and fitting

The authors state, that this makes up to 1000 ways on how to proceed in doing a EVA and therefore brings a great opportunity of making "bad" choices. In the following is described, how to do the VWLS (Variate - Weibull - Least Squares) method, where none of these decisions need to be taken, which makes it more objective than other methods.

I.A. Choosing a distribution

When it comes to extreme value distributions Gumbel, Frénet or Weibull often used for fitting data. These 3 can be generalized by the same function - the GEV.

I.A.1. The Generalized Extreme Values (GEV) Distribution

The cumulative distribution function selected for the VWLS approach is

$$F(x, \vec{\theta} = \mu, \alpha, \gamma) = exp\left(-\left(1 + \gamma\frac{x-\mu}{\alpha}\right)^{-\frac{1}{\gamma}}\right), \quad (1)$$

where α is the scale parameter, μ the location parameter, γ the shape parameter and x the measured data value.

I.B. The plotting positions

Then the so called Weibull plotting position

$$P_m = m/(N+1), \quad (2)$$

Equations are part of the text and need appropriate punctuation

from order statistics is performed, where each data point is assigned a rank m in ascending order $1 \leq m \leq N$, with N the total number of data points. In other words, one sets $F(\xi_m) = P(x_m)$, which is illustrated in fig.1.

To keep track of the variables, ξ denotes the rank position regarding the CDF function F and the measured x corresponds to the probability P_m.

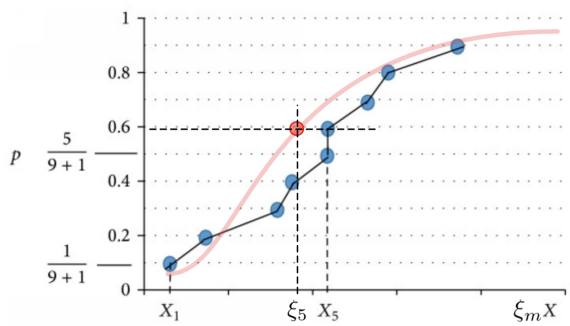

FIG. 1: Ranking data according to Weibull plotting positions (eq. 2). The smooth red line marks the fitted CDF $F(\xi_m)$. Original image[2] modified by Authors.

I.C. Weighing and fitting

In conventional methods, fitting procedures minimize the error regarding the Probability axis P. One of the key elements in the success of the VWLS method is, that it minimizes the error with respect to the variate X axis, using a weighted least squares method,

$$min\left(\sum_{m=k}^{N} w_m (x_m - \xi_m)^2\right), \quad (3)$$

with $w_m = \frac{1}{\sigma_m^2}$ and $\sum_{m=k}^{N} w_m = 1$ (k denotes the smallest indices used for the fitting procedure).

We are beginning the fitting process with

1. setting all σ_m equal and setting initial fitting parameters $\vec{\theta} = \mu, \alpha, \gamma$ for F. Then a iterative procedure starts - this takes between 10-20 iterations:
2. Minimizing with least squares (eq.3)
3. Rejecting data (see sec.I.C.1)
4. Calculating weights (see sec.I.C.2)
5. Back to step 2. until weights and F do not change anymore.

I.C.1. Rejection of data

Smallest extremes are excluded from the analysis (set weights=0), if they are too near to the asymptotic limit. This can be decided by

$$\frac{\xi_{k+1} - \xi_k}{\xi_N - \xi_k} < b, b = 0.001 \tag{4}$$

where ξ_k is the smallest extreme, ξ_N the biggest and works in a iterative process, as long as the equation is met. If the dataset shrinks down to a size < 5, the data is considered unreliable.

I.C.2. Weights

Calculating weights

$$\sigma_m^2 = \int (\xi_m - \bar{\xi}_m)^2 f_m(\xi_m) d\xi_m \tag{5}$$

is done with the mean

$$\bar{\xi}_m = \int \xi_m f_m(\xi_m) d\xi_m,$$

the pdf of the variate X_m

$$f_m(\xi_m) = g(m, N) F(\xi_m)^{m-1} (1 - F(\xi_m))^{N-m} f(\xi_m)$$

with $g(m, N) = \frac{N!}{(m-1)!(N-m)!}$ and the pdf of the GEV CDF

$$f(x) = \frac{dF(x)}{dx}.$$

II. PERFORMANCE

II.A. Evaluation of a fitting method

The journal evaluates the fitting method directly on the probability scale.The probability bins are defined according to the Weibull plotting positions (Eq. 2). For each rank m, the corresponding value of the variable scale is

$$x_m = F^{-1}(P_m).$$

As per the journal, if the fitted distribution is correct, a new random observation should fall into each of the $N + 1$ intervals with equal probability $\frac{1}{N+1}$.

Using Monte Carlo simulations, the relative frequencies $n_{j,r}/n$ of observations falling into each bin are calculated. The deviation from the ideal uniform distribution is measured by

$$C = \sqrt{\sum_{r=1}^{N+1} \left(\frac{n_{j,r}}{n} - \frac{1}{N+1} \right)^2}.$$

, where j denotes the fitting method and r the probability bin. A smaller value of C means better probabilistic performance of the fitting method.

II.B. Comparison of EVA methods

In this section, the VWLS method is compared with common EVA methods like Maximum Likelihood (ML), Probability Weighted Moments (PWM), L-moments and LH2 data.For this, Monte Carlo simulations with 100,000 repetitions were performed using data generated from Fréchet ($\gamma = 0.3$), reversed Weibull ($\gamma = -0.3$) and Gumbel ($\gamma = 0$) distributions with $\alpha = 10$ and $\mu = 70$.

However,the results show that VWLS produces smaller C values than ML, PWM, L and LH2 for almost all sample sizes N. At least,the improvement is especially clear for small samples (e.g. $N = 10$ or $N = 15$).

In extrapolation to x_{60}, the estimation of the mean

$$x_{30/31} = F^{-1}\left(\frac{30}{31}\right)$$

from samples of size $N = 15$ is more accurate with VWLS. The median of the estimated return level is closest to the true distribution compared to the other methods.

III. CONCLUSION

The VWLS method combines the GEV distribution, the Weibull plotting position (Eq.2) and a variance-based weighted least squares minimization. The probability-based evaluation shows that VWLS provides considerably better estimates than ML, PWM, L-moments and LH2 methods, especially for small sample sizes and extrapolated return levels. It also requires no subjective methodological choices. However, the method assumes independent, identically distributed data and it does not account for correlation between order statistics.In spite of limitations,VWLS outperforms other methods,especially for small datasets and for extreme return levels.

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[1]. T. M. Makkonen, L., "An improved method of extreme value analysis." Journal of Hydrology X **2, 100012.** (2018), 10.1016/j.hydroa.2018.100012.

[2]. P. M. Makkonen, L., "Defining sample quantiles by the true rank probability." Journal of Probability and Statistics , 1–6 (2014).

[3] L. Makkonen, "Plotting positions in extreme value analysis." Journal of Applied Meteorology and Climatology **45(2)**, 334–340 (2006).

Presentation —
- On slide 2 you did a nice job with covering the general outline and the problem, but I also felt it was hard to connect the EV analysis approach with a specific problem. For example, you showed rain/moisture plot(s) later on, but it wasn't clear to me if this was for single instance extreme storms, or annual rainfall extremes, or some rainfall rate extremes.
- It might seem trivial, but it wasn't clear whether the rainfall examples/plots were related to predictions or post-dictions; which I'm sure was clear to you, but not to the audience.
- Slide 6 had a lot of content, and was slightly distracting.
- The choice of having the table on slide 7 was good, and that gave the audience a chance to really listen and focus on the speaker, and to not be distracted by trying to understanding all the various elements that might appear on the slide.
- Starting to better explain the method itself on slide 8–which is ~50% of the way through the presentation—left me pretty confused, because you were showing results from the VWLS without having gone through it too much beforehand. On slide 5 you showed that it's the "Variance Weibull Least Squares", but that doesn't explain the methodology that well, which leaves the audience having to remember the impact of the VWLS (from a few minutes earlier) while the VWLS method is being described in more detail.
- It was well done covering the CDF estimate, and then the minimization, and then the EV loss. It made it easier to understand the methodology approach.
- Good job answering questions.

Presentation : 4/5

Write-up —
- The topic was very much appropriate, and much of the write-up was organized to explain the VWLS approach.
- There were gaps in explanation where it seemed you felt compelled to summarize the paper, but connections to the method or context were lost for the reader, i.e. me.

Write-up : 4/5